

SUJAL DHAKAL

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EDUCATION

Georgia Institute of Technology

Expected May 2027

Bachelor of Science in Computer Science

Atlanta, GA

- **GPA:** 3.9
- **Concentrations:** Artificial Intelligence & Human-Centered Systems
- **Related Coursework:** Data Structures & Algorithms, Databases & SQL, Algorithm Design, Computer Architecture & Programming, User Interface Design, Software Design & UML, Object-Oriented Programming, Computing in Python, Discrete Mathematics, Linear Algebra

EXPERIENCE

Bits of Good - Hack4Impact Chapter

Jan. 2025 – Present

Engineering Manager

Atlanta, GA

- Orchestrated development of medical distribution platform for **56 facilities** and **100,000+** annual items shipped.
- Automated distribution workflows, reducing manual efforts by **80%** and accelerating inventory lookups from **hours to seconds**, by introducing Azure AI Services with OpenAI models and RAG pipelines.
- Authored **60+ GitHub issues**, reviewed **400+ commits**, and resolved merge conflicts to a team of 5 developers.

Georgia Institute of Technology

Aug. 2024 – Present

Undergraduate Researcher

Atlanta, GA

- Collected **2.4 million** health metrics from a **4-week study** on **12** Dementia patients to track agitation events.
- Cleaned datapoints to **≥90% completeness** by converting Firebase real-time data to CSVs for Pandas processing.
- Modeled agitation prediction with a **semi-supervised LSTM**, reaching **AUROC [0.87]** on multimodal wearable streams, by applying anomaly detection to partially labeled data using PyTorch and Numpy.

Technique Newspaper

Nov. 2024 – Sept. 2025

Software Developer

Atlanta, GA

- **Cut page load by 8s** by replacing WordPress front-end with static-rendered Next.js for Tech's student newspaper.
- Rebuilt UI/UX, increasing **Lighthouse Accessibility and SEO to 96/100**, with TailwindCSS and React.
- **Reduced API load by 55%** by adding middleware that caches WordPress REST calls and merges frequent fetches.

PROJECTS

Intl. Children's Advisory Network | TypeScript, Next.js, MongoDB, TailwindCSS, Zod

Jan. 2025 - Aug. 2025

- Built **9-step** medication scheduler for **1,000 children** with modular components, state logic, and Zod validation.
- Improved backend maintainability in **10 services** by implementing Mongoose models and service modularization.
- Implemented **JWT-based password reset system** in Node.js with token expiry to safeguard brute force attacks.

DataFly CLI - SQL Data Generation | Go, Cobra, SQL, LangChain

May 2025 - Present

- Decreased test environment setup time by **85%** by generating relational SQL data in **under 2 minutes**.
- Built a **LangChain-based LLM parser** to convert SQL schemas into faker files in YAML, supporting 4 SQL dialects.
- Ensured foreign key integrity across complex schemas by building **relationship-aware data generators**.
- **Cut runtime load by 30%** using chaining on schema parsing and generation to avoid redundant processing.

Personality Academy | JavaScript, Next.JS, Prisma, PostgreSQL, CSS, ImageKit, GCP

Sept. 2022 - Aug. 2024

- Launched full-stack learning platform for **3,000+ users** across 23 chapters, tracked via completion analytics.
- Delivered sub-second **multi-filter search** for **2,000+ profiles**, grouping 512 categories via optimized SQL joins.
- Built management for **2,000 profiles + 13,000 content items**, replacing spreadsheets and saving **10+ hrs/week**.
- Cut image load times from **12s to 2s** with ImageKit CDN, sustaining **99.9% uptime** during traffic spikes.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, Go, Swift, HTML/CSS, C++, Assembly

Frameworks & Tools: React, Next.js, Node.js, Flask, Django, React Native, Express, TailwindCSS, Zod, Prisma

ML & Data: PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, LangChain, RAG, OpenCV

Dev & Tech: Git, Docker, Google Cloud, Azure, Firebase, MongoDB, Claude Code, Vercel, Jupyter Notebook